

# **Student Utility Program using Java Programming Fundamentals**

## **Objective / Aim**

To understand and implement the basic programming constructs of Java, including variables, data types, operators, conditional statements, loops, and methods by developing a menu-driven Student Utility Program.

## **Algorithm / Logic Description**

### **Part A – Student Information**

1. Start the program.
2. Display a menu.
3. Accept student name, roll number, and marks.
4. Calculate total marks.
5. Calculate percentage.
6. Display all student details.

### **Part B – Conditional Statements**

1. Read a number.
2. Check whether it is Even or Odd.
3. Read three numbers.
4. Find the largest number.
5. Display grade using if-else.
6. Display the day of the week using switch.

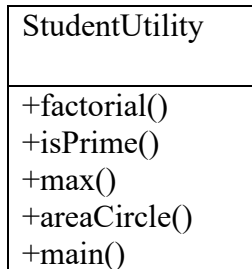
### **Part C – Looping Statements**

1. Read a number N.
2. Print multiplication table.
3. Print numbers from 1 to N.
4. Calculate sum of first N natural numbers.
5. Generate Fibonacci series.

### **Part D – Methods**

1. Create methods for factorial, prime check, maximum of two numbers, and area of a circle.
2. Call each method from main().
3. Display the returned results.
4. End the program.

### UML Class Diagram



### Source Code :

```
package tasks;

import java.util.Scanner;

public class StudentUtilityProgram {

    int factorial(int g){

        int fact=1;

        for(int i=1;i<=g;i++){

            fact=fact*i;

        }

        return fact;

    }

    double areaOfCircle(double radius){

        return 3.14 * radius * radius;

    }

    boolean isPrime(int j){

        if(j <= 1){
```

```

        return false;
    }
    for(int i = 2; i <= j / 2; i++){
        if(j % i == 0){
            return false;
        }
    }
    return true;
}

int maximum(int a, int b){
    if(a > b){
        return a;
    }
    else{
        return b;
    }
}

public static void main(String[] args) {
    StudentUtilityProgram su=new StudentUtilityProgram();
    Scanner sc = new Scanner(System.in);
    int choice;
    do {

        System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
        System.out.println("1. Student Information");
        System.out.println("2. Even or Odd");
        System.out.println("3. Largest of Three Numbers");
    }
}

```

```
System.out.println("4. Grade");
System.out.println("5. Day of Week");
System.out.println("6. numbers from 1toN");
System.out.println("7. sum of n natural numbers");
System.out.println("8. multiplication table");
System.out.println("9. fibonacci");
System.out.println("10. Factorial");
System.out.println("11. Prime Number");
System.out.println("12. Maximum of Two Numbers");
System.out.println("13. Area of Circle");
System.out.println("14. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
switch(choice) {
    case 1:
        String name = "Mohana";
        int roll = 101;
        int marks1 = 80;
        int marks2 = 90;
        int marks3 = 85;
        int total = marks1 + marks2 + marks3;
        double percentage = total / 3.0;

        System.out.println("Name: " + name);
        System.out.println("Roll Number: " + roll);
        System.out.println("Total: " + total);
        System.out.println("Percentage: " + percentage);
```

```
break;
```

```
case 2:
```

```
int number= sc.nextInt();
```

```
if(number % 2 == 0){
```

```
    System.out.println("Even");
```

```
}
```

```
else{
```

```
    System.out.println("Odd");
```

```
}
```

```
break;
```

```
case 3:
```

```
int a=sc.nextInt();
```

```
int b= sc.nextInt();
```

```
int c=sc.nextInt();
```

```
if(a>b && a>c){
```

```
    System.out.println(a);
```

```
}
```

```
else if(b>c){
```

```
    System.out.println(b);
```

```
}
```

```
else{
```

```
    System.out.println(c);
```

```
}
```

```
break;
```

```
case 4:
```

```
int per= sc.nextInt();
```

```
if(per>=90){
```

```
        System.out.println("Grade A");
    }
    else if(per>=75){
        System.out.println("Grade B");
    }
    else if(per>=50){
        System.out.println("Grade C");
    }
    else {
        System.out.println("Fail");
    }
    break;
case 5:
    int day= sc.nextInt();
    switch(day){
        case 1:
            System.out.println("Monday");
            break;
        case 2:
            System.out.println("Tuesday");
            break;
        case 3:
            System.out.println("wednesday");
            break;
        case 4:
            System.out.println("Thursday");
            break;
```

case 5:

System.out.println("friday");

break;

case 6:

System.out.println("saturday");

break;

case 7:

System.out.println("sunday");

break;

default:

System.out.println("Invalid");

}

break;

case 6:

int l= sc.nextInt();

for (int i=1;i<=l;i++){

System.out.println(i);

}

break;

case 7:

int n= sc.nextInt();

int sum=0;

for(int i=1;i<=n;i++){

sum=sum+i;

}

System.out.println(sum);

break;

case 8:

```
int num= sc.nextInt();  
for(int i=1;i<=10;i++){  
    System.out.println(num+" x "+i+" = "+num*i);  
}  
break;
```

case 9:

```
int k=sc.nextInt();  
int d=0,e=1;  
for(int i=1;i<=k;i++){  
    System.out.print(d+" ");  
    int f=d+e;  
    d=e;  
    e=f;  
}  
break;
```

case 10:

```
System.out.println(su.factorial(5));  
break;
```

case 11:

```
System.out.println(su.isPrime(5));  
break;
```

case 12:

```
System.out.println(su.maximum(5,6));  
break;
```

case 13:

```
System.out.println(su.areaOfCircle(2.3));
```



```

        break;
    case 14:
        System.out.println("Thank You");
        break;
    default:
        System.out.println("Invalid Choice");
    }
} while(choice != 14);
sc.close();
}
}

```

### Output :

```

===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade
5. Day of Week
6. numbers from 1toN
7. sum of n natural numbers
8. multiplication table
9. fibonacci
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice: 1
Name: Mohana
Roll Number: 101
Total: 255
Percentage: 85.0

===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade
5. Day of Week
6. numbers from 1toN

```

```

===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade
5. Day of Week
6. numbers from 1toN
7. sum of n natural numbers
8. multiplication table
9. fibonacci
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice: 14
Thank You

Process finished with exit code 0

```